Hemostasis

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Learning Objectives:

1. Describe the steps that occur in primary and secondary hemostasis
2. Differentiate between the intrinsic, extrinsic, and common pathways of the coagulation cascade
3. Identify mechanisms of anticoagulation that prevent hypercoagulability states
4. Describe fibrinolysis as the final step in hemostasis
5. Apply the principle of hemostasis to clinical scenarios

Source: CPG

Why is hemostasis important?

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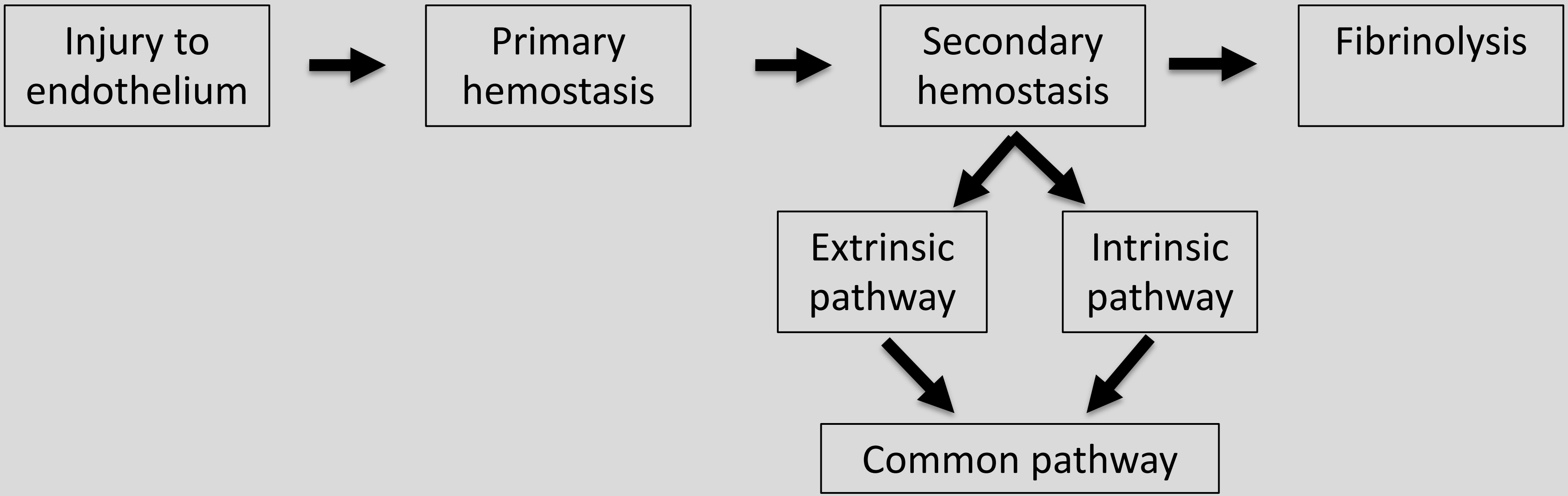
Hemostasis

- Physiological process of stopping a bleed caused by damage to the endothelium



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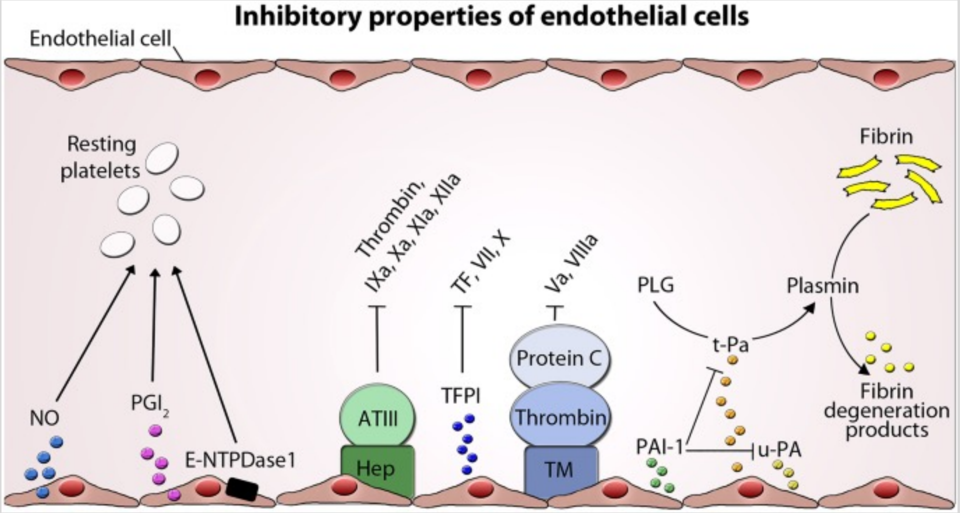
Endothelium

Normal= antithrombotic

- Antiplatelet adhesion- NO, PGI2, nucleoside diphosphatase
- Anticoagulation- heparin like molecules, tissue factor pathway inhibitor, thrombomodulin
- Fibrinolytics- tPA



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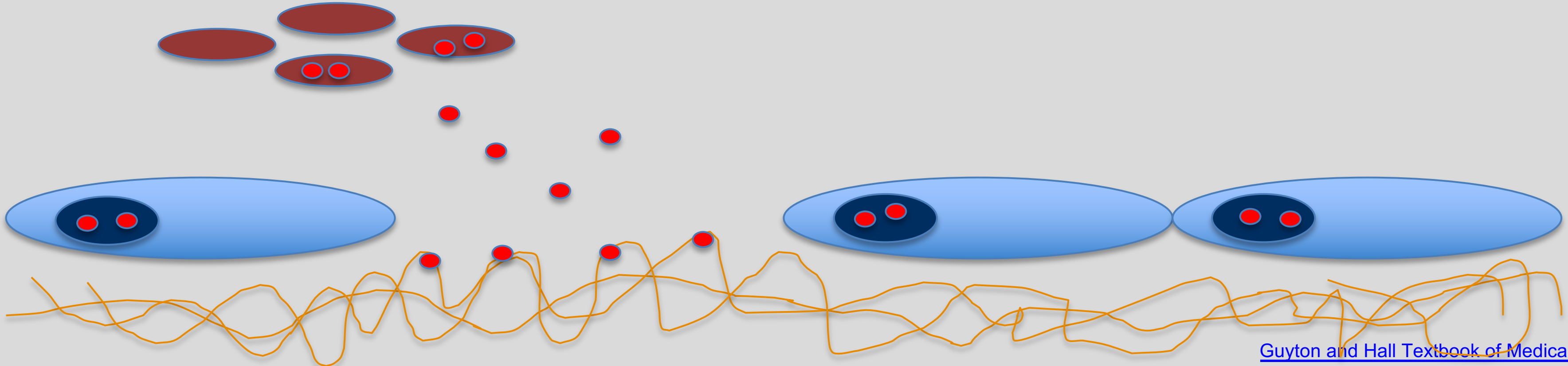
Endothelium

Injury= prothrombotic

- Damaged endothelial cells
  - Endothelin→ causes vasoconstriction
- Exposure of subendothelial collagen
  - vWF binds to collagen
- Von Willebrand Factor (vWF)
  - Circulating plasma protein
  - Platelets (alpha granules)
  - Endothelial cells (Weibel-Palade bodies)



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Primary hemostasis

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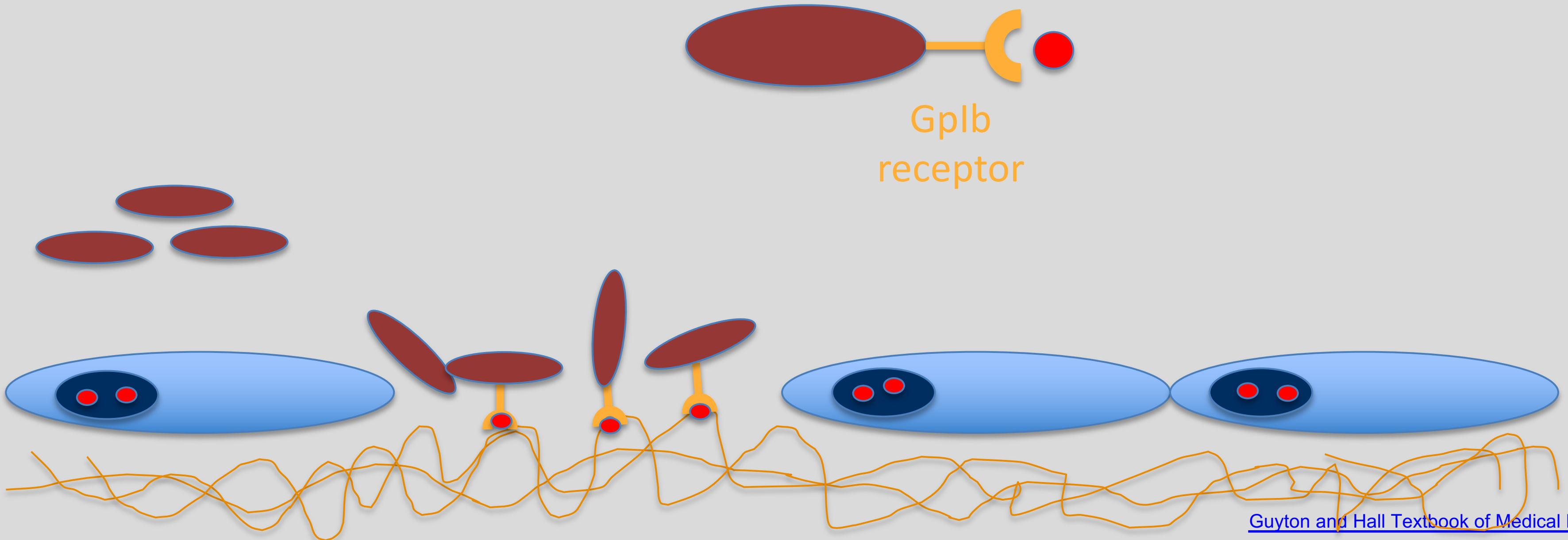
Formation of the platelet plug

- Adhesion
  - vWF binds to GpIb (platelet surface receptor)



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Primary hemostasis

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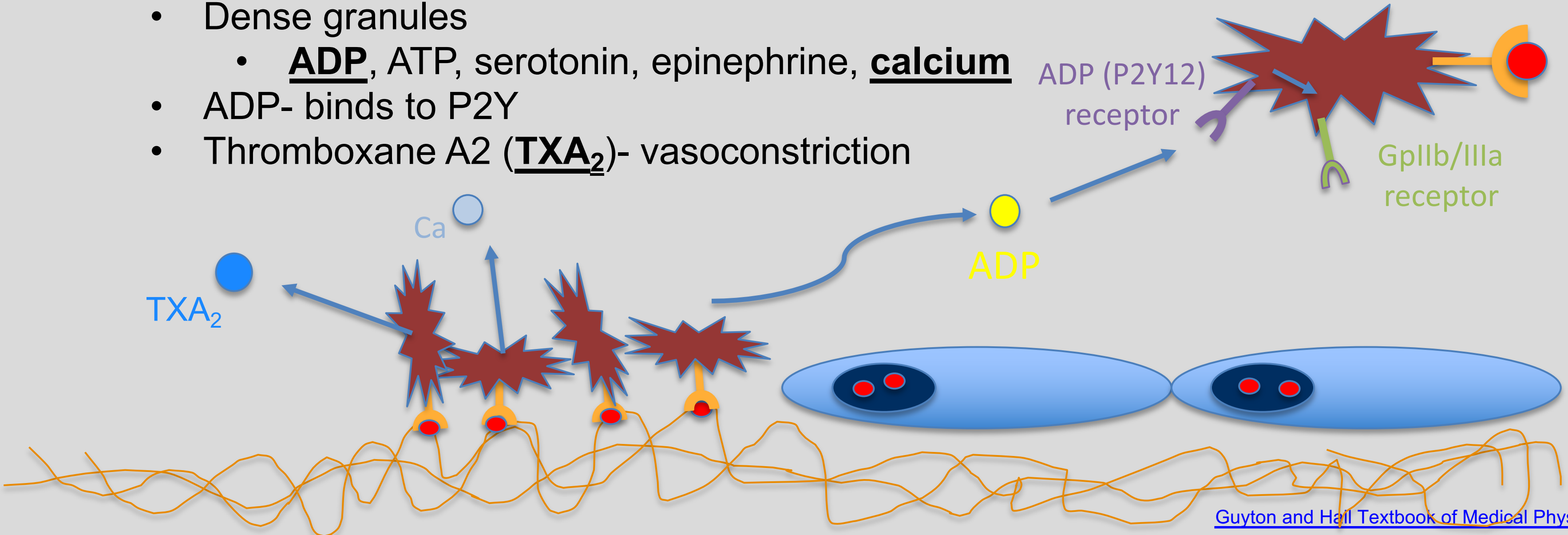
- Activation
  - Platelets change shape and release positive feedback mediators
    - Alpha granules
      - P-selectin, fibrinogen, factor V, **vWF**, fibronectin, platelet factor 4, PDGF, TGF-B
    - Dense granules
      - **ADP**, ATP, serotonin, epinephrine, **calcium**
    - ADP- binds to P2Y
    - Thromboxane A2 (**TXA_2**)- vasoconstriction



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Primary hemostasis

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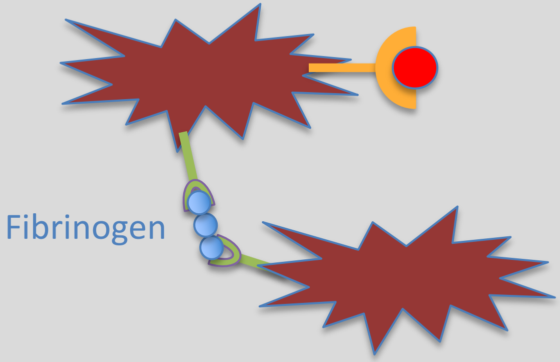
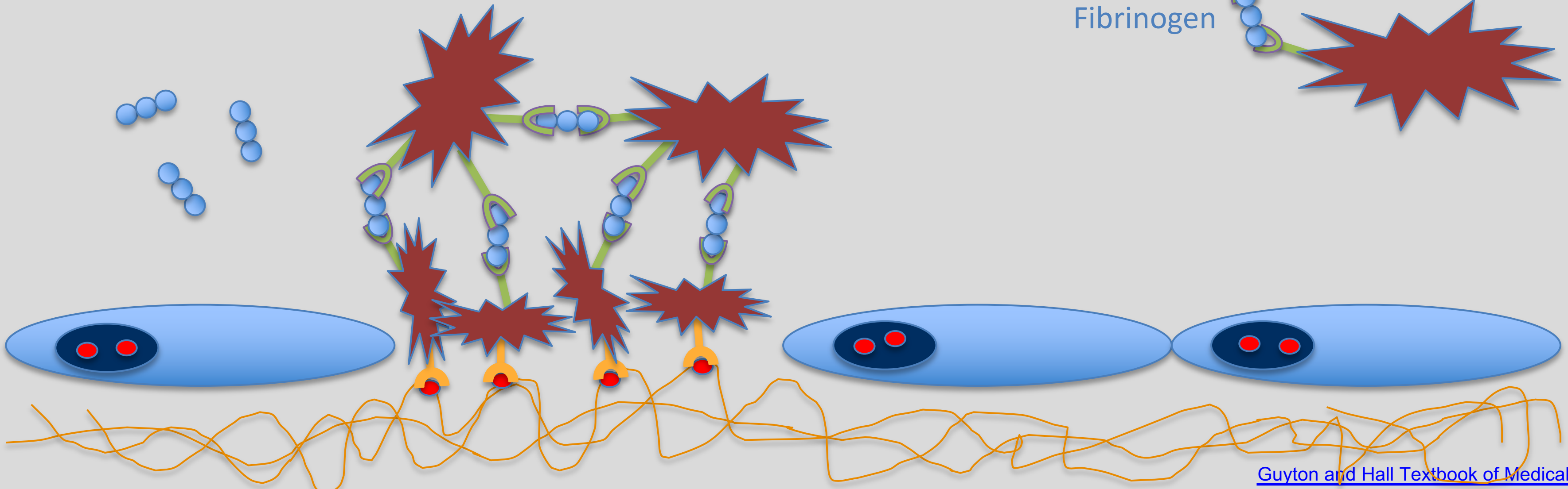
- Aggregation
  - GpIIbIIIa receptor binds fibrinogen from plasma
  - Platelets aggregate and form platelet plug





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Question

Which of the following has the correct receptor and ligand pair?

a) GpIb receptor and endothelin
b) P2Y12 receptor and Von Willebrand Factor
c) P2Y12 receptor and calcium
d) GpIIbIIIa receptor and fibrin
e) GpIIbIIIa receptor and fibrinogen

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Secondary hemostasis

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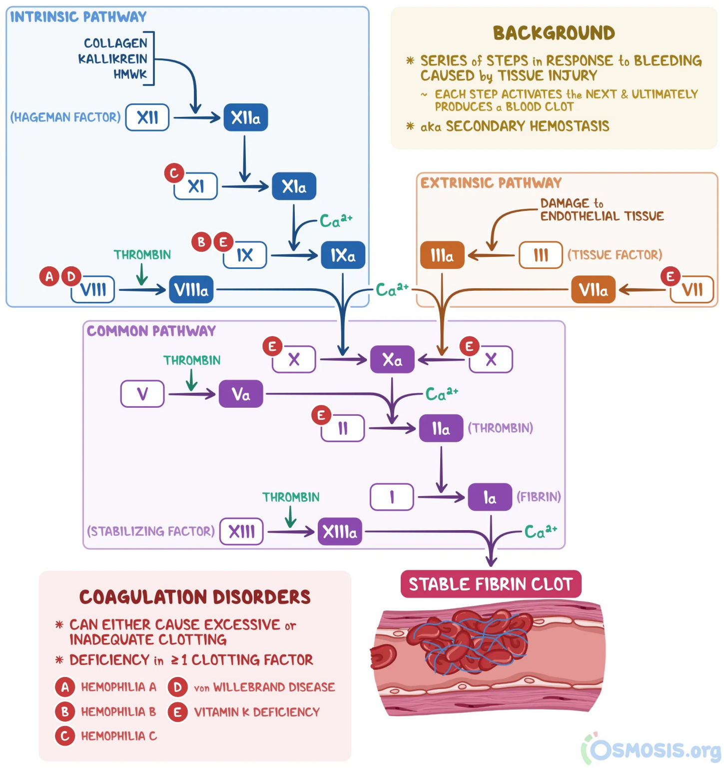
Coagulation cascade→ stabilization of the platelet plug

- Fibrinogen→ fibrin (more stable)



Osmosis: Coagulation Cascade

Secondary hemostasis

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Intrinsic pathway- damage inside vasculature

- Contact activated
  - Factor XII interacts with negatively charged collagen, kallikrein, and kininogen (HMWK) and is activated

XIIa

$\downarrow$

XI → XIa

$\downarrow$ Ca

IX → IXa

VIII $\xrightarrow{\text{Thrombin}}$ $\downarrow$ Ca

X → Xa

- Thought to be slower and generate more robust response
- **aPTT**= activated partial thromboplastin time
  - Normal= 25-40 seconds
  - Used to monitor heparin therapy

Important factors= XII, XI, IX, VIII

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Secondary hemostasis

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Extrinsic pathway- endothelial damage, causes blood to leave vasculature

- Tissue factor (factor III) is expressed on subendothelial muscle cells and fibroblasts

- Quicker than intrinsic pathway
- Factor VII is a Vit K dependent factor produced by the liver
- **aPT**= activated prothrombin time
    - Normal= 11-15 seconds
    - Used to monitor warfarin therapy

  Important factors= III, VII

IIIa ← III

Ca ↓ VIIa ← VII

TF-FVIIa

↓

Xa ← X

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Secondary hemostasis

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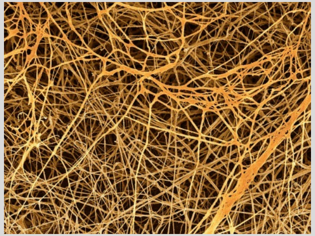
Common pathway- both intrinsic and extrinsic pathways lead to Factor Xa

- Thrombin:
  - Promotes platelet aggregation and activation
  - TXA_2 synthesis
  - Stimulates endothelial cells to produce adhesion molecules and GF
  - Has anticoagulant effect on healthy endothelium
- Retraction
  - Platelets contract, fibrin mesh shrinks, brings edges of damaged vessel together



Science Photo

Xa

V → Va ↓ Ca

II (prothrombin) → IIa (thrombin)

↓

I (fibrinogen) → Ia (insoluble fibrin)

Ca ↓ XIIIa $\xleftarrow{\text{Thrombin}}$ XIII

Crosslinked fibrin

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Question

A young girl falls and scraps her knee. After the platelet plug is formed, what pathway will be activated next to stabilize the platelet plug?

a) Primary hemostasis
b) Intrinsic pathway
c) Extrinsic pathway
d) Common pathway
e) Fibrinolysis

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Anticoagulation checks

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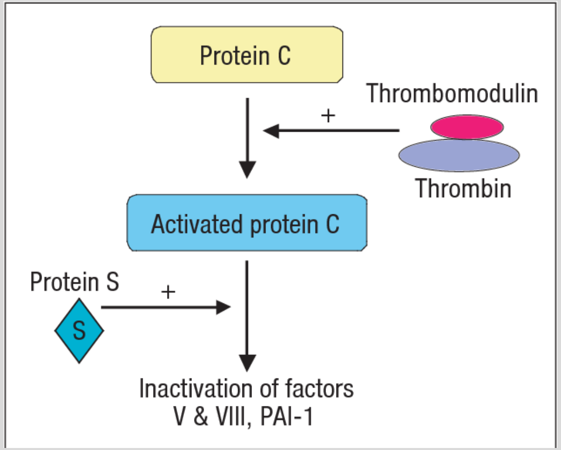
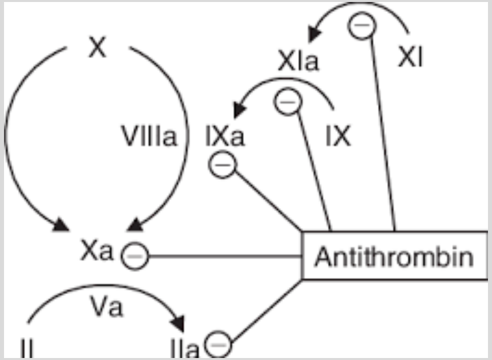
Regulation of clot formation

- Protein C and S- Vit K dependent synthesis in the liver
    - APC complex binds to thrombin and thrombomodulin inhibits factors Va and VIIIa
- Antithrombin- binds and degrades thrombin, factor IXa and Xa



Photo



Photo

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Fibrinolysis

Degradation of the fibrin network by plasmin

tPA, Kallikrein,
Streptokinase,
Urokinase

Plasminogen → Plasmin → Fibrin mesh

Fibrin mesh ↓ Soluble fibrin

Question

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What is the main role of antithrombin?

a) Positive feedback on the coagulation cascade
b) Bind and activate thrombin
c) Bind and degrade thrombin
d) Bind thrombin, protein C and protein S to inactivate clotting factors
e) Bind fibrinogen and convert it to fibrin

Question

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Disorders of hemostasis

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Primary hemostasis:
Von Willebrand disease- deficiency of vWF
Bernard-Soulier syndrome- deficiency of GpIb receptors
Glanzmann thrombasthenia- deficiency of GpIIb-IIIa receptors

Secondary hemostasis:
Hemophilia- low levels of clotting factors, X-linked recessive

- Hemophilia A- lack of factor VIII
- Hemophilia B- lack of factor IX

Clinical Significance

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Labs:
Decreased platelet number and increased bleeding time= thrombocytopenia
Normal platelet number and increased bleeding time= think platelet dysfunction
Elevated PT/INR or aPTT= think clotting factor disorder

- Elevated PT/INR= extrinsic pathway
  - Factor VII deficiency
- Elevated aPTT= intrinsic pathway
  - Hemophilia- factors VIII, IX, XI
- Elevated PT/INR and aPTT= extrinsic and intrinsic pathways
  - Deficiency of vitamin K-dependent pathway
  - Factors II, V, VIII, X

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Clinical Significance

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Drugs:

Antiplatelets: prevent platelet plug formation

- Aspirin– inhibits COX-1→ is required to make thromboxane in platelets= decreased platelet aggregation
- Clopidogrel- binds ADP (P2Y12) receptors→ inhibits activation of GPIIb/IIIa receptor complex= decreased platelet aggregation

Anticoagulants: prevent synthesis of clotting factors

- Heparin- binds to ATIII→ increases affinity for factor Xa and thrombin→ binds and inactivates them
    - Monitored with aPTT
- Warfarin- inhibits VKORC1→ depletes functional vitamin K→ depletion of factors II, VII, IX, X
    - Inhibits Protein C and S
    - Monitored with aPT/INR

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Question

A 50-year-old man comes into the ED complaining of black stools for the past week. When you ask about his diet, he tells you that he is a very picky eater and eats a lot cereal, grains and potatoes. He does not eat much of meat, fruits or vegetables. You think that he may have a vitamin deficiency that is making him bleed. Which of the following has been depleted due to his poor diet?

a) Platelets
b) Von Willebrand Factor
c) Calcium
d) Factors V and X
e) Factors II and VII

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Vit K dependent= factors II, VII, IX, X, protein C and S

Summary

1. Injury
2. Primary hemostasis
    1. Vasoconstriction
    2. Platelet plug
        1. Adhesion
        2. Activation
        3. Aggregation
3. Secondary hemostasis
    1. Intrinsic vs extrinsic
    2. Common pathway
4. Fibrinolysis

Core References

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Guyton and Hall Textbook of Medical Physiology, Fourteenth Edition
Chapter 37- Hemostasis and Blood Coagulation
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Osmosis: Coagulation Cascade
https://www.osmosis.org/answers/coagulation-cascade

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